

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: CIV102

Course: ENGINEERING MECHANICS

QP No. :U020R-1

Duration: 3 hours

Max. Marks:100

PART -A

Answer all the questions

$10 \times 2 = 20$ Marks

1. A block of mass 30 kg is placed on a smooth plane which is inclined at 30° with horizontal. The block is restrained from sliding down by a force F acting parallel to the plane. Determine the magnitude of the force F .
2. A string of length 20 cm is attached to a point A on a smooth vertical wall and to a point B on the surface of a sphere of radius 10 cm as shown in the Figure 1. The sphere, whose weight is 300 N, hangs in equilibrium against the wall. Find the tension in the string and the reaction of the wall.

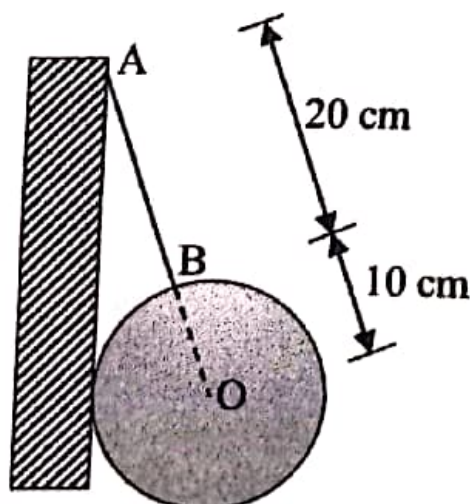


Figure 1

3. A force of 100 kN acts along a line joining two points O (0, 0, 0) and A (3, 4, 5) m. The component of this force in the x-direction is _____.
- (a) 57.6 kN (b) 42.4 kN (c) 100 kN (d) 142.4 kN

4. The centroidal y distance for the equilateral triangle shown in Figure 2 is _____.

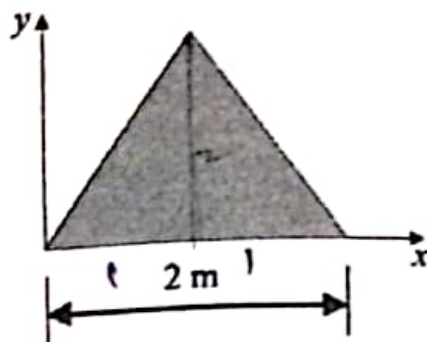


Figure 2

5. Determine the polar moment of inertia of the shaded area given in the Figure 3.

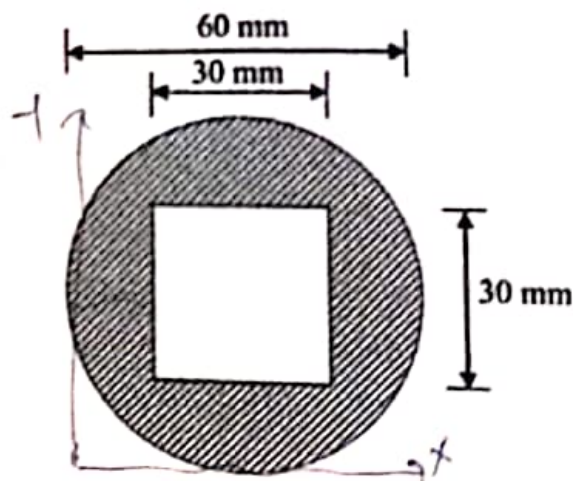


Figure 3

6. Determine the volume generated when the shaded quarter circular area is revolved by 270° about the y-y axis as shown in the Figure 4.

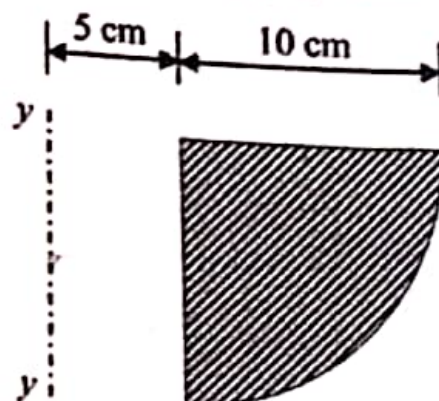


Figure 4

7. Find the inclination α of the block A shown in Figure 5 at which it will start to slip down if the coefficient of friction between the block and the surface is 0.24.

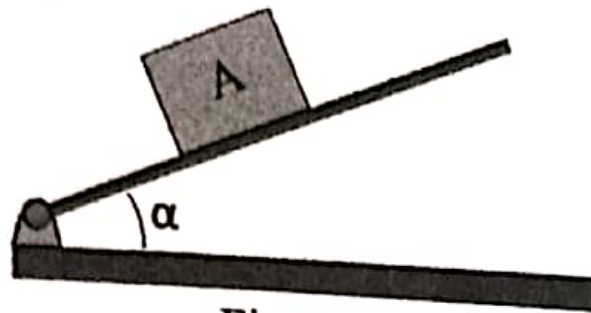


Figure 5

8. Block A and B shown in the Figure 6 are of masses 30 kg and 70 kg respectively. The blocks are connected with an inextensible cable that passes over a pulley as shown. Determine the minimum value of coefficient of static friction required to maintain equilibrium of the system.

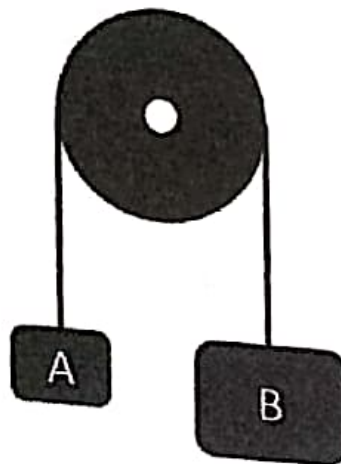


Figure 6

9. A particle is thrown vertically upwards with a velocity of 49.05 m/s. What will be the velocity of the particle after 3 seconds?
10. A car moving with a velocity of 30 km/h has a kinetic energy of 52000 J. When the velocity of the car is increased to 60 km/h, the work done is _____.

PART - B

Answer all the questions

$4 \times 15 = 60$ Marks

11. Four people are attempting to move a stage platform across the floor. If they exert the horizontal forces as shown in the Figure 7, determine:

- (a) equivalent force-couple system at O and,
 (b) the points on the x- and y- axes through which the line of action of the single resultant force R passes.

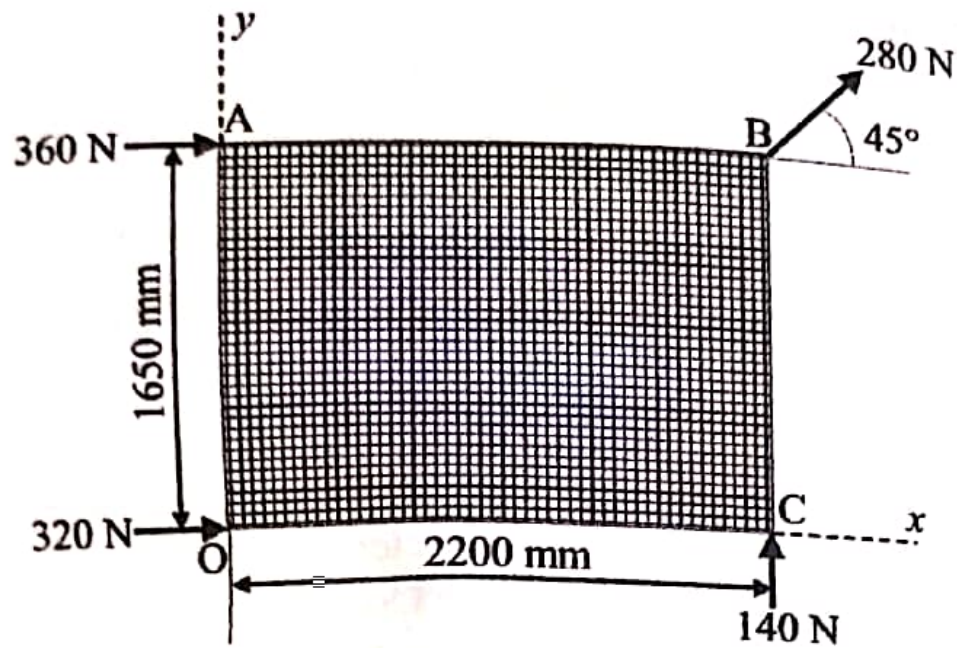


Figure 7
 (OR)

12. Three cables are joined at D. Due to an applied force of 60 kN at D, the cables are aligned as shown in Figure 8. The applied force passes through a point E on the ground as shown. The two ends A and B are on the wall and the other end C is on the ground. The wire CD is vertical. Find the forces in all the three wires.

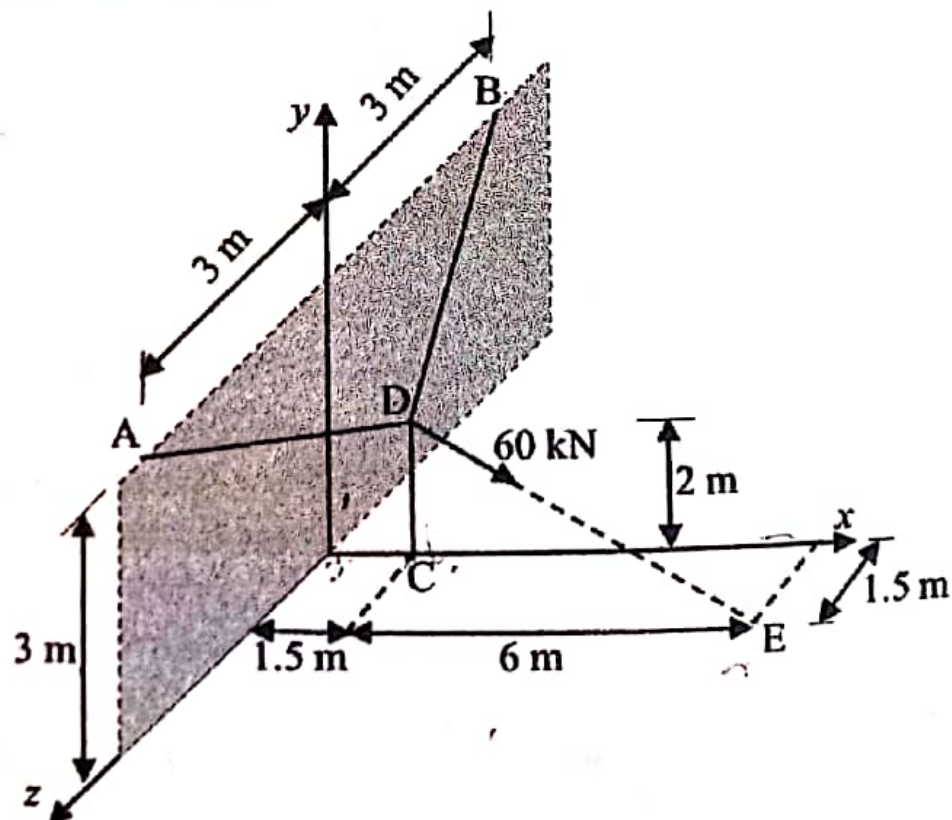
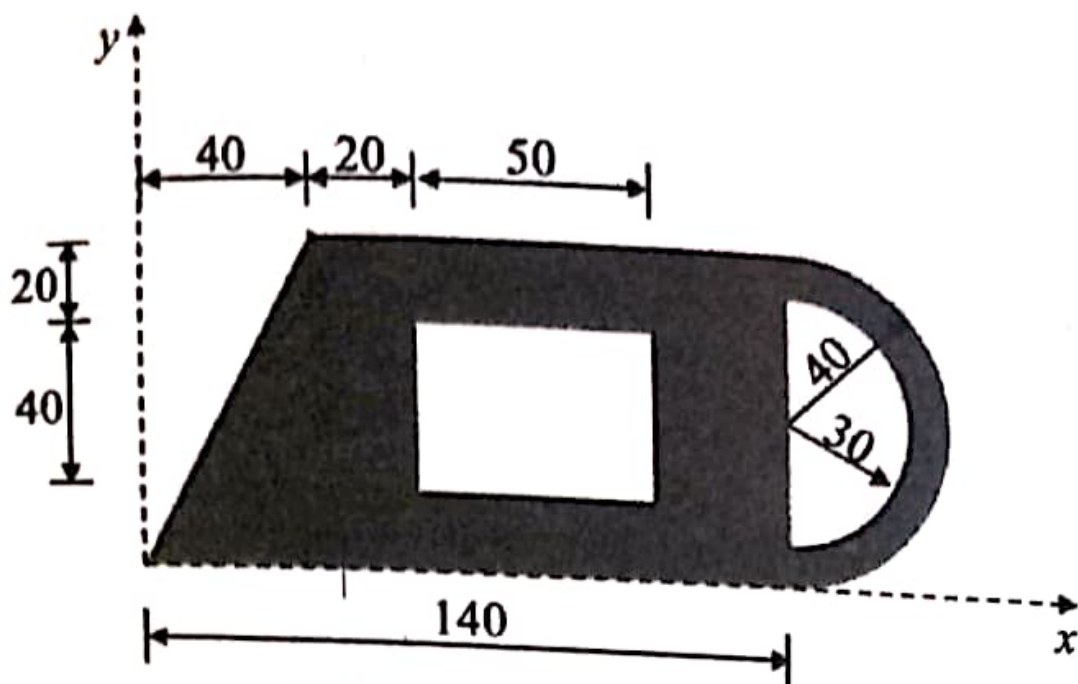


Figure 8

13. Determine the coordinates of the centroid of the shaded area given in the Figure 9.



All dimensions are in mm

Figure 9

(OR)

14. Determine the radius of gyration of the shaded area given in the Figure 10 about the centroidal x-x and y-y axes.

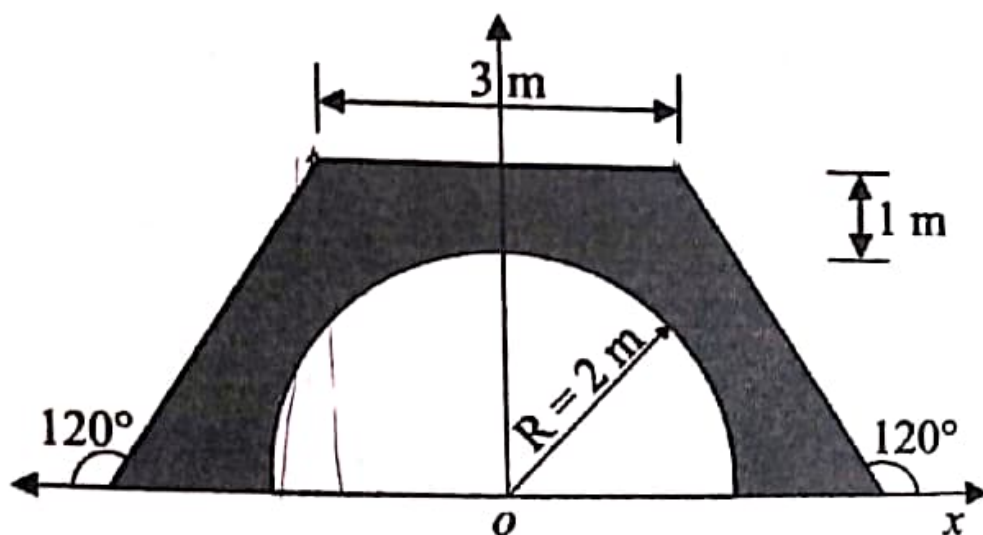


Figure 10

15. Block A of mass 30 kg rests on block B of mass 40 kg as shown in Figure 11. Block A is restrained from moving by a horizontal rope tied at point C. Determine the minimum force P applied, parallel to the plane which is inclined at 30° with horizontal, necessary to move the block B down the plane. Take coefficient of friction for all contact surfaces as 0.35.

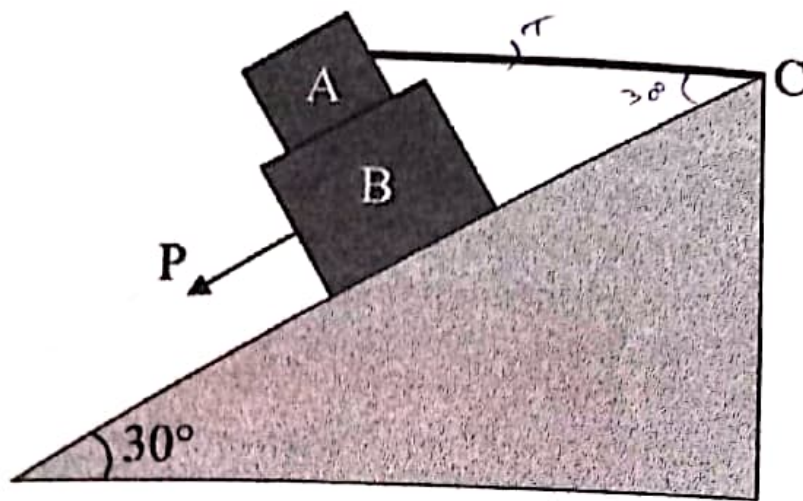


Figure 11
(OR)

16. The ladder shown in Figure 12 is of length 6 m and is supported by a horizontal floor and a vertical wall. The coefficient of friction between the floor and the ladder is 0.4 and between the wall the ladder is 0.25. The weight of the ladder is 200 N and it supports a vertical load of 900 N at C as shown. Determine the least value of α at which the ladder may be placed without slipping. Determine the reactions at that stage.

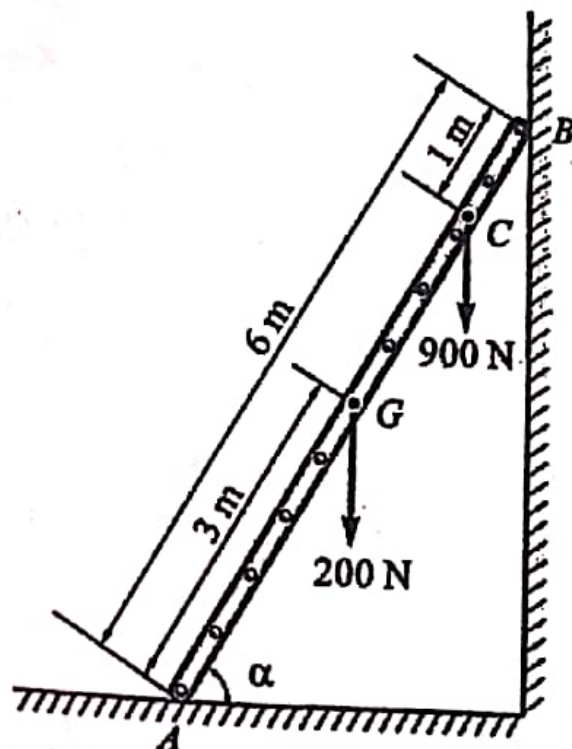


Figure 12

17. Two objects A and B are dropped from the top of a tower of height h at different time intervals and reach the ground in t_1 seconds and t_2 seconds, respectively. Find the ratio of t_1 and t_2 , if the object A travels 50 m in its

last second of flight, while object B travels 70 m in its last second of flight.

(OR)

18. Two blocks A and B of masses 100 kg and 50 kg respectively starts from rest. The blocks are connected with inextensible strings passing over frictionless pulleys as shown in the Figure 13. Determine acceleration of each block and tension in the string. Take coefficient of friction between the block A and the floor as 0.25.

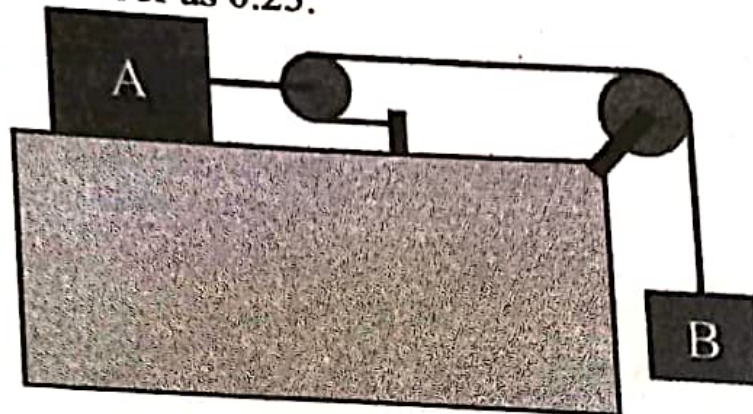


Figure 13

PART - C

Answer the following

$1 \times 20 = 20$ Marks

19. (A) Two smooth cylinders, each of weight equal to 1500 N and diameter of 40 cm, are connected at their centres P and Q by a string of length 50 cm and rest upon a horizontal floor as shown in the Figure 14. This system supports another cylinder with centre R which is of weight 2500 N and diameter 40 cm as shown. Find the tension in the string PQ and the forces developed at the contacts at the floor and between the cylinders.

(10 marks)

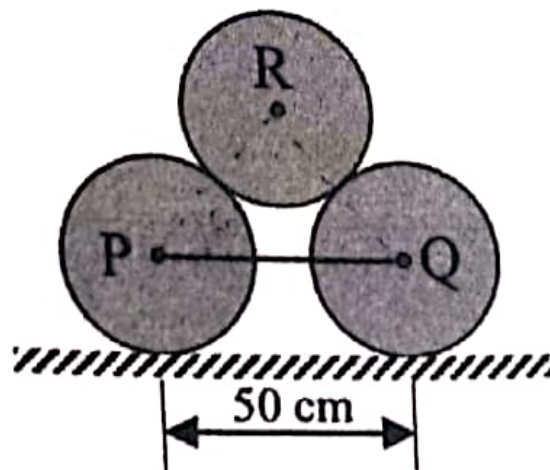


Figure 14

- (B) A basketball player shoots a free throw to a basket 4.6 m away and 3 m off the floor as shown in the Figure 15. If the ball is released from a point 2.1 m above the floor and at an angle of 50° , determine:
- (a) the required initial velocity u ; *0.41*
 - (b) the time the ball passes through the rim; *0.92*
 - (c) the maximum height of the trajectory above the ground; *17.6*
 - (d) the speed of the ball and the angle of its trajectory as it passes through the rim.
- (10 marks)

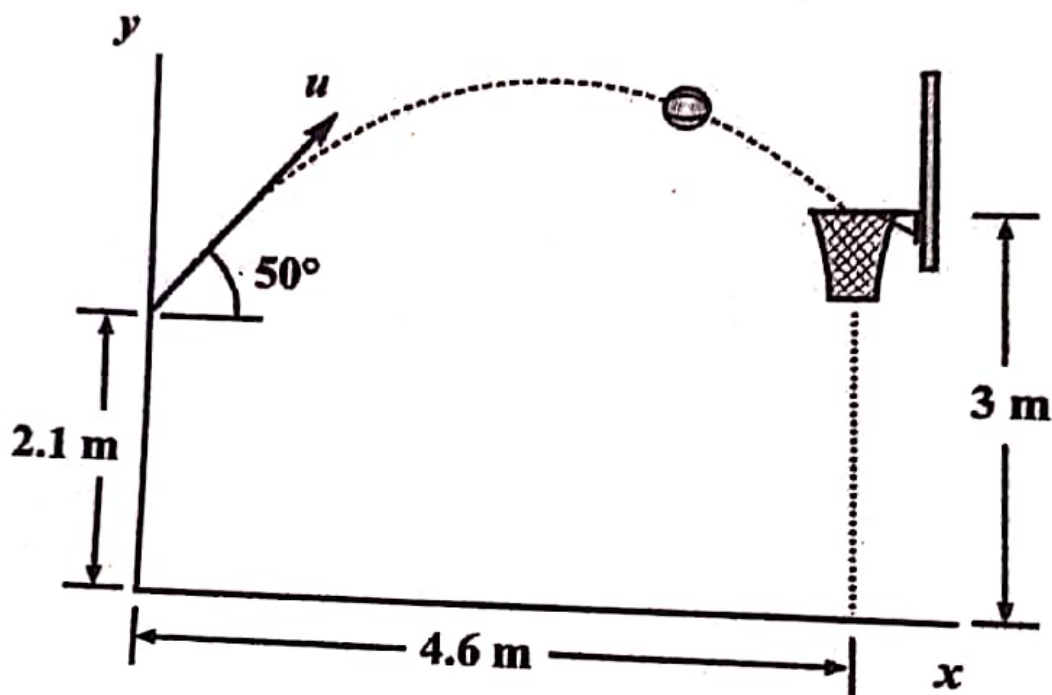


Figure 15
